

Q1

a)

		Weather conditions next day		
		sunny	overcast	Rainy
Weather conditions on present day	Sunny	0-39	40-74	75-99
	Overcast	0-14	15-64	65-99
	rainy	0-24	25-69	70-99

for every day, a two-digit number is chosen from the random number table. The first number is taken from the first row and first column, and each next number is chosen from the subsequent column until all numbers from that row are chosen- then the next row will be used. This number determines the weather conditions the next day, using the table above. For example if it is sunny and the first number is 18, this would be between 0-39 and the next day will be sunny. Then the next two-digit number is picked from the table, which is 69, the present day is sunny, and 69 is between 40-74 so the next day will be overcast.

Date	Weather conditions	Random number	Weather condition next day
Sept 1	Sunny	18	Sunny
Sept 2	sunny	69	Overcast
Sept 3	overcast	84	Rainy
Sept 4	rainy	78	Rainy
Sept 5	rainy	85	Rainy
Sept 6	rainy	78	Rainy
Sept 7	rainy	40	Overcast
Sept 8	overcast	23	Sunny
Sept 9	sunny	95	Rainy
Sept 10	rainy	76	Rainy
Sept 11	rainy	25	

- b) Out of 10 'next days', 5 of the days were unchanged from the previous day, giving a probability of 5/10 or 50%. These days were sept 2, sept 5, sept 6, sept 7, and sept 11. Using this probability of 50%, it can be estimated that during a whole year of 365 days, 182.5 days will have the same weather conditions as the previous day.
- c) There is a possibility of start-up bias, since during the year there is seasonality in weather, meaning there are generally different probabilities of certain weather conditions during certain seasons. Therefore the first table may only be accurate during September, and this may give an inaccurate simulation for the rest of the year, given the start-up probability conditions.
- However, in this simulation there is no 'carrying over' of random digits, so there is no possibility of numbers piling up over time and causing further start-up bias.

Q2

a)

State list			Statistics	
Time (mins)	In queue	Using pump	Waiting time	Lost? Y/N
0.00	-	-	-	-
1.71	-	C1	0.00	N
6.51	C2	C1	0.20	N
6.71	-	C2	0.00	N
7.52	C3	C2	3.22	N
10.62	C3	C2	0.12	Y (C4)
10.74	-	C3	0.00	N
11.23	C5	C3	3.61	N
13.33	C5	C3	1.51	Y (C6)
14.84	-	C5	0.00	N
18.58	C7	C5	0.26	N
18.84		C7	0.00	N
19.04	C8	C7	3.26	N
22.30	-	C8	0.00	N
25.43	C9	C8	0.44	N
25.87	-	C9	0.00	N
30.34	-	-	0.00	N
31.66		C10		
Event list				
R.N	event	Time (mins)	Done? Y/N	
6524	Arrival C1	1.71	Y	
3011	Arrival C2	4.80	N	
7493	Service C1	5.00	Y	
7654	Arrival C3	1.01	N	
5070	Service C2	4.03	Y	
3768	Service C3	3.51	Y	
4608	C4 is lost	3.10	N	
5243	Service C3	4.10	Y	
8595	Arrival C5	0.61	N	
5010	Service C5	4.00	Y	

5921	C6 is lost	2.10	N
2692	Arrival C7	5.25	N
3662	Service C7	3.46	Y
8923	Arrival C8	0.46	N
2024	Arrival C9	6.39	N
3924	Service C8	3.57	Y
6180	Service C9	4.47	Y
2108	Arrival C10	6.23	N

b) Total waiting times = 12.62 mins, ignoring lost customers = 10.99, divided by 9 customers served = average waiting time of 1.22 mins.

Total of 2 customers lost in 30 mins, therefore an expected loss of 4 customers per hour.

Q3

a) $1 - P(3 \text{ or less}) = P(4 \text{ or more})$

$P(3 \text{ or less}) = P(0) + P(1) + P(2) + P(3)$
 $= (2/5)(3/5)^0 + (2/5)(3/5)^1 + (2/5)(3/5)^2 + (2/5)(3/5)^3$
 $= 0.4332$

$P(4 \text{ or more}) = 1 - 0.4332 = 56.68 \%$

b) F

c) i) $p = 3/5$ and $L_q = (0.6^2)/(1-0.6) = 0.6$ machines waiting in the queue to be repaired

ii) $W_q = L_q / \lambda = 0.6 / 3 = 0.2$ days expected waiting time in queue

d) Expected number of machines being repaired at any given time = $5 - 0.6 = 4.4$ machines

Expected cost per day = wage + cost of lost productivity = $120 + 95 * 4.4 = £568$

With two repairpersons, number of machines being repaired doubles to 8.8, and expected machines waiting in the queue is halved to 0.3, which = $240 + 95 * 8.8 = £1028$
 $1028 > 568$, so the costs are higher, which would not be profitable.

e) Triangular distribution = $0.8 < 2.7$ and $0.8 \leq 1.3 \leq 2.7$

f) Assumption 1- Arrivals in an M/G/1 system follow a Poisson process, so inter-arrival time is exponentially distributed

Assumption 2- service times are independent and identically distributed.

It is reasonable to assume that the arrival process is a Poisson process. However, the second assumption may not be realistic as the given service times are triangularly distributed, which would mean that the probability distribution is not the same for each machine which violated the assumption of an M/G/1 queue.